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Computer Architecture

1. Different Between input, output, Memory and Processing units with examples

unit	function	Examples
Input	Accepts the data and Instruction from the user or extranal devices	Keyboard, mouse, scanner, microphone
output	sends Processed data to the users or other devices	monitor, printers speakers
memory	stress data instruction and intermedists ~~so~~ results tempealty or permenently	RAM, ROM, Harddisk, cache
Processing unit	Perform arthemetic, logic, control and decision making operations	ALU, controLunit Register

2. A Program, executs 10^8 instruction on a Processor with a clock cycle time of 1ns and an average (P) of 2 calcutate the execution time if another Processor execute the same Program with a (P) 1.5 and clock cycle time of 0.8 ns, find the speed ups.

Sol:

Given

Program instructions : 10^8

Processor1 : c(P) = 2 , clock cycle time = 1ns

Processor2 : c(P) = 1.5 clock cycle time = 0.8ns

Execution time = Ic × c(P) × clock cycle time.

Processor 1:

$T_1 = 10^8 \times 2 \times 1ns = 2 10^8 ns - 0.25$

Processor 2:

$T_2 = 10^8 \times 1.5 \times 0.8ns = 1.2 \times 10^8 ns = 0.125$

Speed up:

Speed up $= T_1/T_2 = 0.2 / 0.12 = 1.667$

So, Processor 2 is $\sim 1.67 \times$ faster

3. DIB hand wired and microprogrammed control which is faster and why?

features	Hardwired control	micro Programmed
Implement	uses fixed logic circuit	uses memory with microcontrol
flexibility	less for complex Instructions faster	Lower, easier to design slower
complexity	High for complex Instruction	High flexiable easy to update or modify
speed	faster	slower
Example	ALU control	CPU with microprogra-mmed instruction set

Hardwired control is faster because it generates control singnals using logic circuits without the need to fetch micro instructions from memory

A. Write the micro operation for the instruction Add R_1 R_2 Assume a register - register architecture

- instruction : Add R_1, R_2 → $R_1 = R_1 + R_2$
- micro operation

1. $AR \leftarrow R_1$ (load R_1 into the Arithmetic Register)
2. $BR \rightarrow R_2$ (load R_2 into another temporary Register)
3. $ALU \rightarrow AR$ (Perform addition is ALU)
4. $R_1 \rightarrow ALU$ (store result back to R_1)

5. Show the timing diagram for the following instruction sequence in a 5-stage pipeline

I_1 : LW R_1, 0(R_2)

I_2 : ADD R_3, R_1, R_4

I_3 : SUB R_5, R_3, R_6

a) without forwarding

b) with forwarding

instruction

I_1 : LW R_1, 0(R_2)

I_2 : ADD R_3, R_1, R_4

I_3 : SUB R_5, R_3, R_6

PIPELINE STAGES: IF, ID, EX, MEM, WB

a) without Forwarding: stass with occurs Becauses IR depends on R_1 from I_1 and I_3 depand on R_3 from I_2

b. with forwarding: Data harzards resolved by flowarding path, fewer stalls

6. Explain supercaler operation, How is it diffiently from pipelining?

super scaler operation:

- Execute multiple instruction in a single cc using multiple functional units
- Hardware detector inpendent instructions and issus them simultaneously

Eg, Inter core i7 can issue 4 instructions per cycle